CSE400B CAPSTONE PROJECT

Deep Learning based Multimodal Emotion recognition from Audio & Text

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PROJECT OVERVIEW

Speech is a natural medium for expressing emotions, making Speech Emotion Recognition (SER) an important tool for human-computer interaction. While traditional SER systems focused on audio features, they often struggled with noise, speaker variability, and limited feature selection. Advances in deep learning have improved performance, but relying only on audio still misses critical emotional cues from text. To address this gap, our goal is to develop a multimodal SER system using the IEMOCAP dataset, combining audio features with BERT-based text embeddings to achieve higher accuracy and real-world applicability in healthcare, virtual assistants, and customer service.

AIM

To develop a multimodal Speech Emotion Recognition (SER) system by integrating audio (MFCC features) and text (BERT embeddings) using the IEMOCAP dataset, to improve emotion classification accuracy compared to unimodal models.

PROBLEM STATEMENT

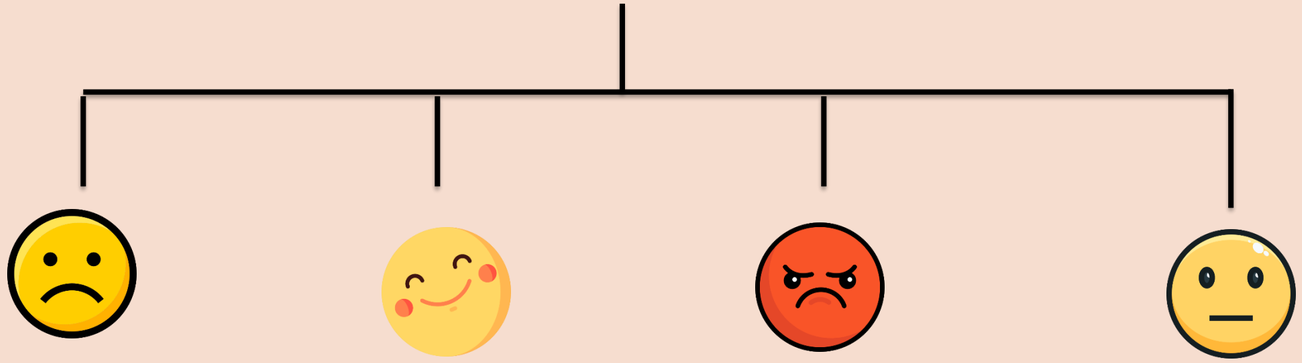
Most existing SER systems rely only on audio, which makes them vulnerable due to noise and speaker variation. Research shows that combining audio with text improves accuracy, but such multimodal approaches remain underused. This project addresses the gap by developing a multimodal SER model on the IEMOCAP dataset to achieve better performance.

OBJECTIVE

- To investigate and evaluate various feature extraction techniques for audio-based emotion recognition and determine the most effective approach for capturing emotional characteristics from speech signals.
- To build a deep learning model for audio features and text embeddings for emotion recognition.
- To implement and facilitate comparison between fusion methods for combining audio and text in a multimodal SER system.
- To create a user-friendly system for real-time emotion prediction.

DATASET OVERVIEW

IEMOCAP dataset



OVERALL EMOTION COUNT

Emotion	Count
Neutral	1708
Happiness	1636
Anger	1103
Sadness	1084

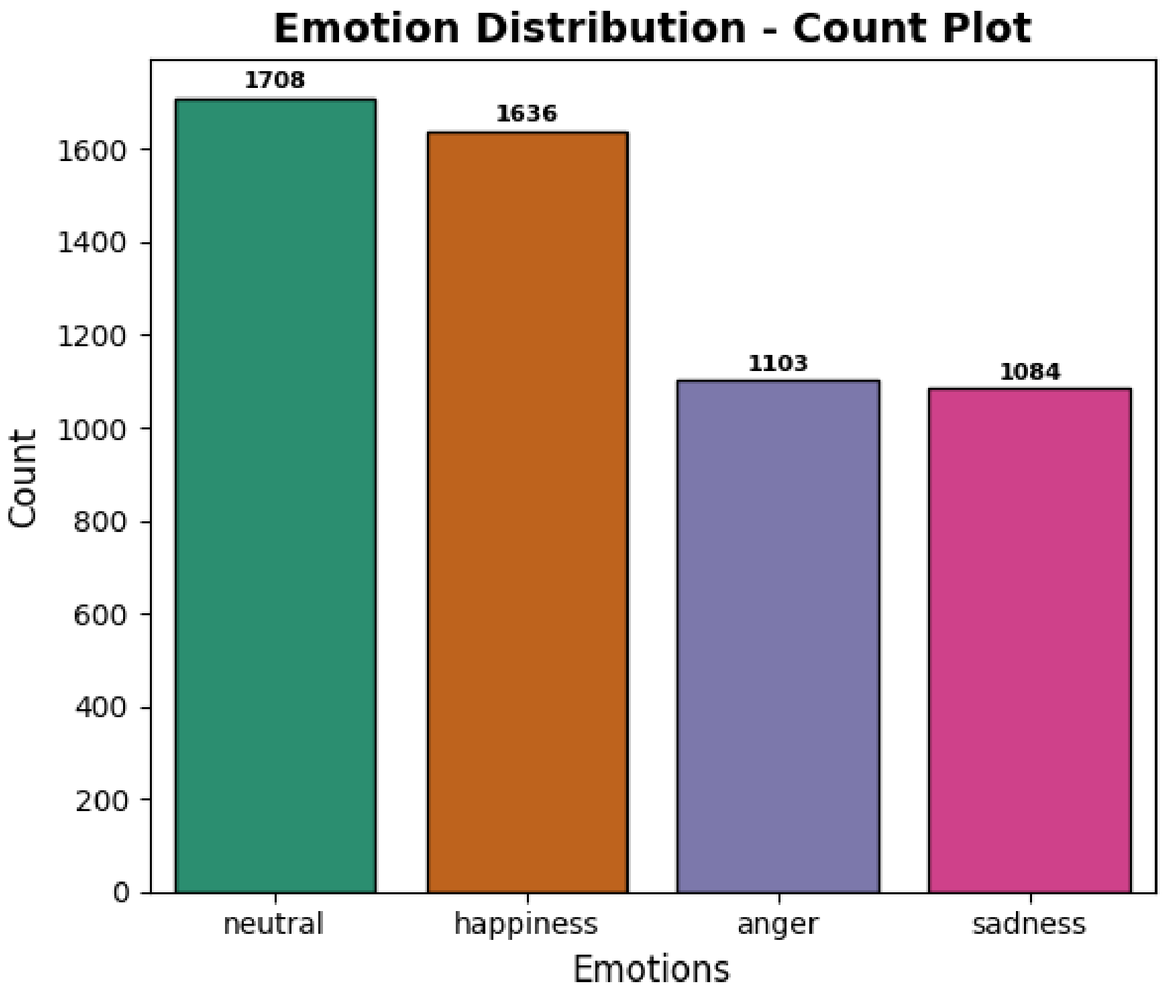
🖼️ **Total dataset size: 5,531**

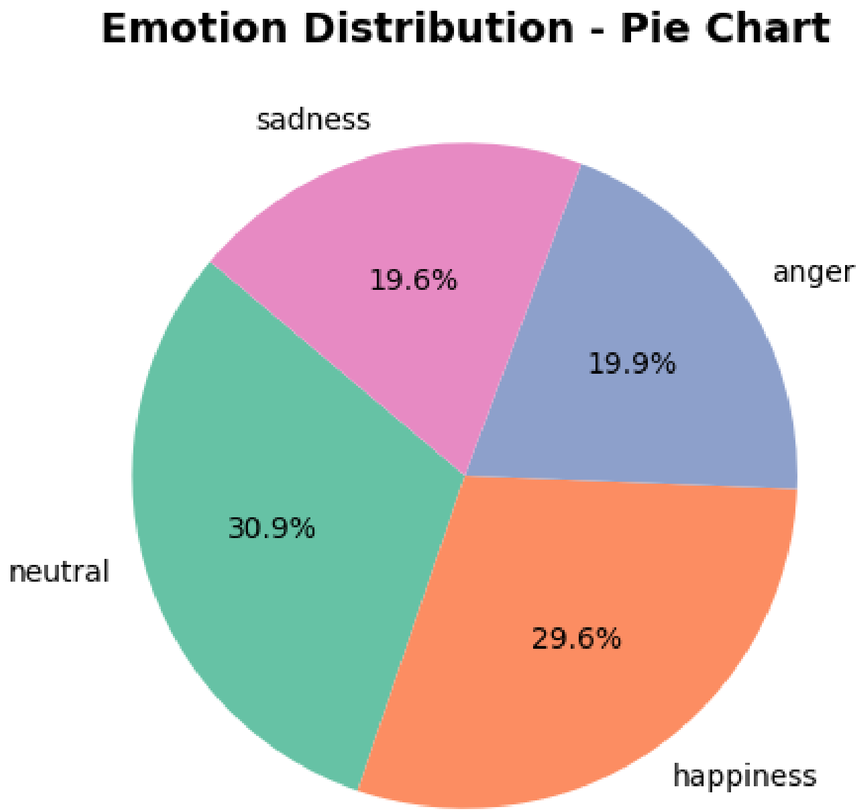
📈 **Most common emotion: Neutral (1,708 samples)**

📉 **Least common emotion: Sadness (1,084 samples)**

⚖️ **Imbalance ratio: 1.58 : 1**

Class Distribution





AUDIO MODALITY

Session ID, Filepath, .wav, utterances

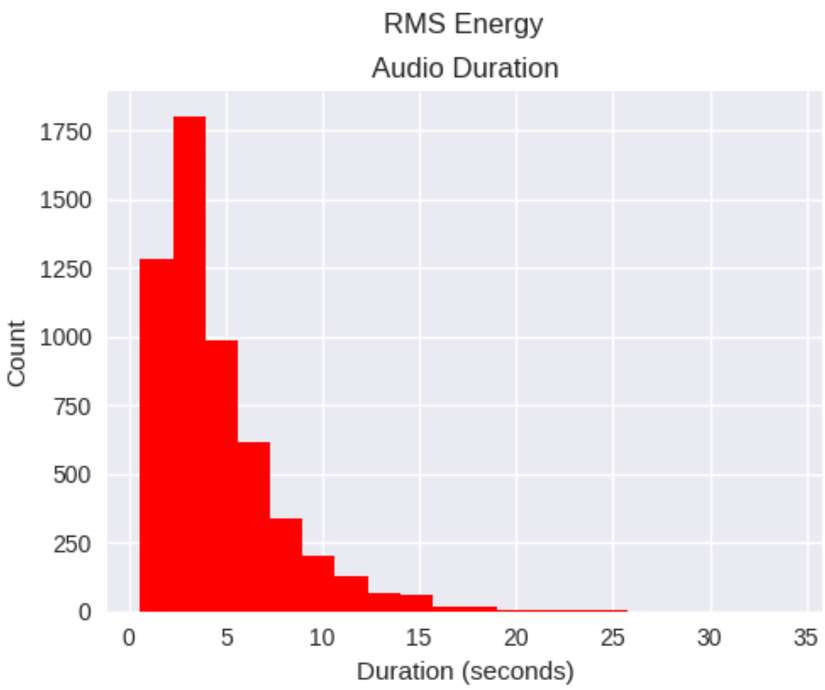
Sampling rate 16000

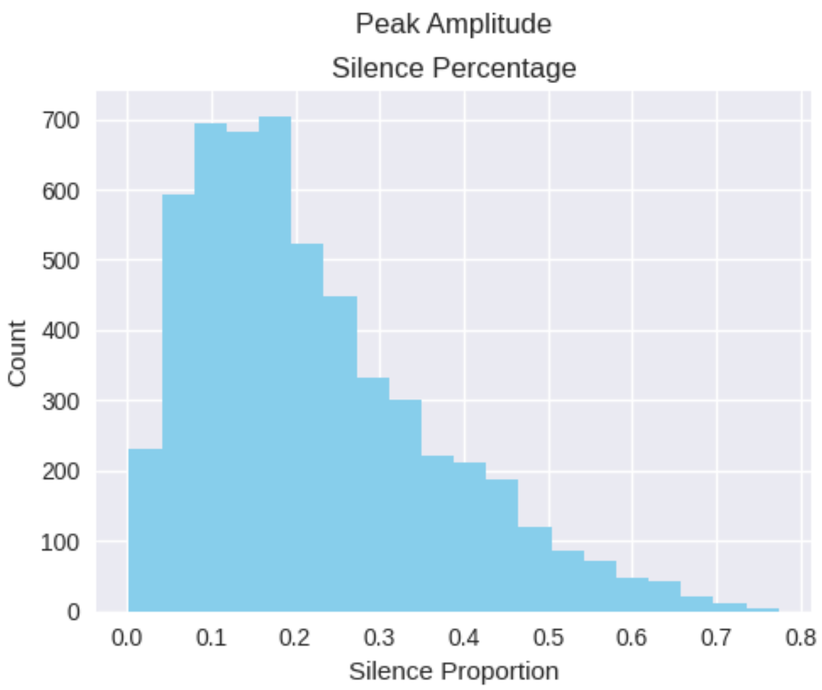
Mono audio

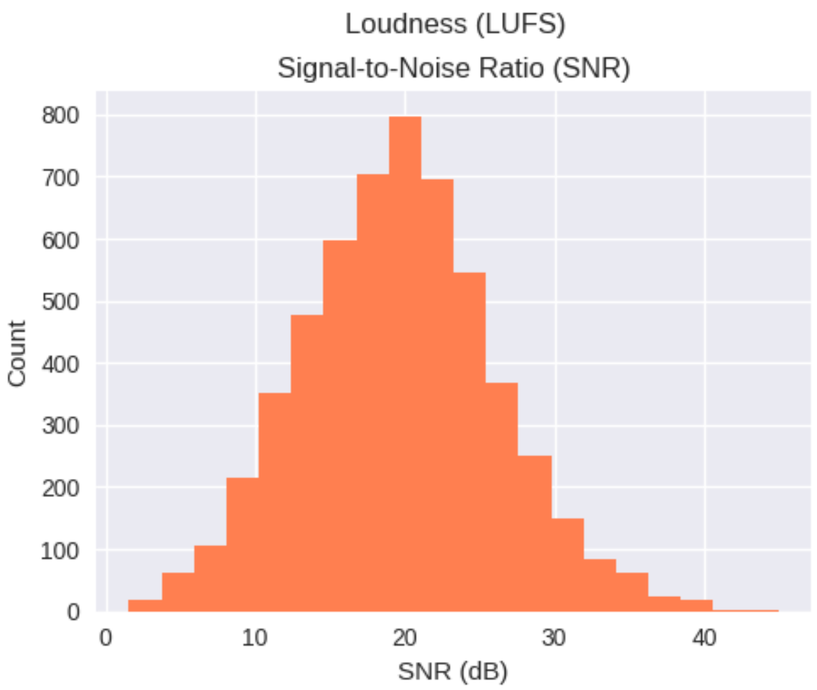
No broken/ invalid files

Audio Level statistics

Audio level statistics





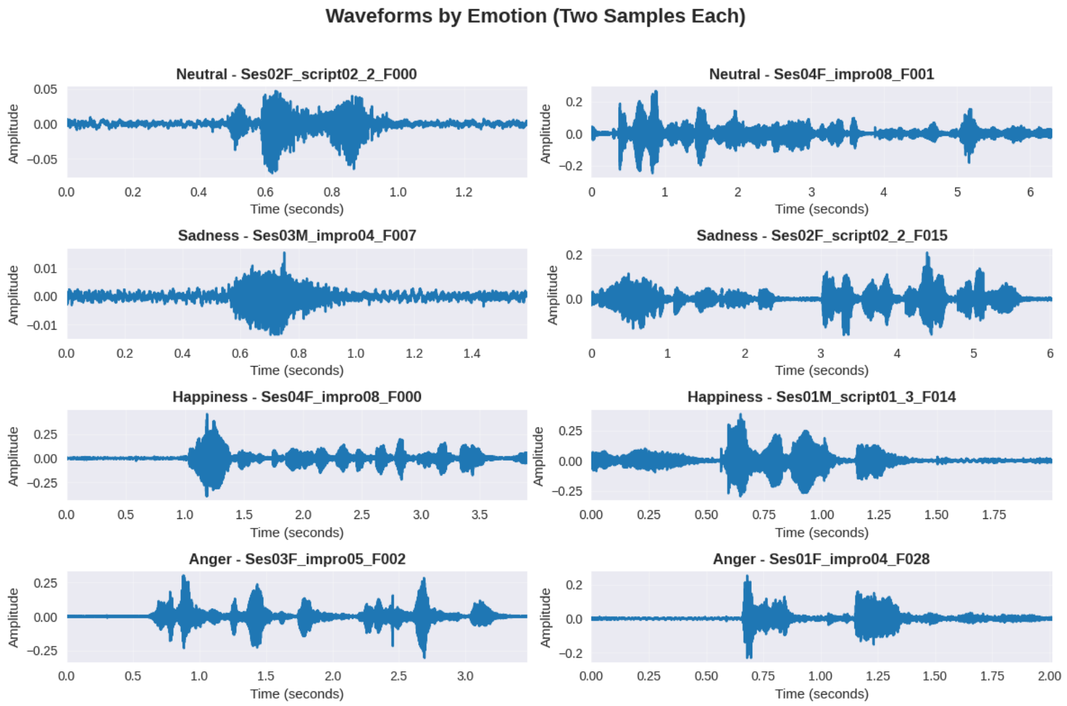


Audio Duration, Silence Percentage, and SNR (LUFS) capture speaking patterns and recording quality, which can indicate emotional state (e.g. hesitation, low energy, or calmness).

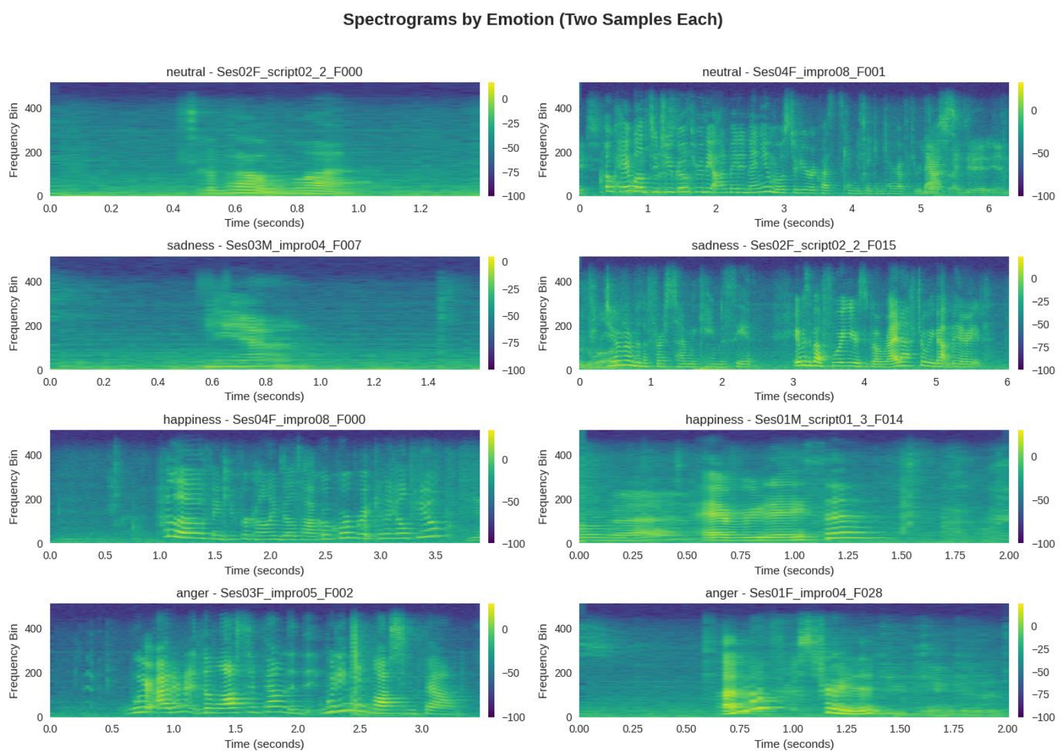
RMS Energy, Peak Amplitude, and Loudness (LUFS) reflect vocal intensity and dynamics of speech, such as excitement (high energy) or sadness (low energy).

Sampled random audio files

WAVEFORM ANALYSIS



SPECTROGRAM ANALYSIS



Design and Implementations

CUSTOM DATASET CLASS

- ✓ Loaded the audio from the dataframe
- ✓ Truncate/padding
- ✓ For the specified feature, extraction will be done.
- ✓ Delta Feature will be applied if specified
- ✓ Depending on features, dimensions will be calculated.

Design and Implementations

will be
ified
features,
ll be

SPILT & DATALOADERS

- ✓ The dataset was splited into 70:15:15.
- ✓ With a batch size 32, the extracted features were loaded into train/test/validation loader; having shuffling only on train dataset

Custom
- ✓ four
layer
- ✓ Atter
- ✓ Glob
- ✓ Class
(Relu
- ✓ Two
outp

Design and Implementations

MODEL SETUP

Custom CNN

- ✓ four Convolutional Feature Extraction layers
- ✓ Attention Layer (SE)
- ✓ Global Pooling Layer
- ✓ Classifier with activation function (Relu) & dropout rate of 0.3
- ✓ Two fully connected layers lead to an output layer (emotion classes)

Custom LSTM

- ✓ Bidirectional LSTM Encoder with 2 layers
- ✓ Hidden Dimension = 128
- ✓ Attention Mechanism
- ✓ Classification Head with batch normalization along with relu activation to reduce overfitting and convert into logits for output classes

Design and Implementations

HYBRID: COMBINING THE CNN AND LSTM MODELS

- ✓ All 4-layer Convolutional layers were used to extract features from inputs like a spectrogram.
- ✓ Channel attention from the CNN architecture
- ✓ Reshape Output for LSTM — From 2D Feature Map to Temporal Sequence
- ✓ Dynamic LSTM Initialization
- ✓ BiLSTM extends LSTM by capturing both past and future temporal dependencies in audio.
- ✓ The final prediction to output classes were on tasked in classification layer.

Design and Implementations

MODEL TRAIN & EVALUATION

- ✓ Training Function on AMP for faster computation.
- ✓ Applied Early Stopping to avoid overfitting.
- ✓ Model Evalutation function which will take the trained model results
- ✓ Trained all three models separately on MFCC and Log-Mel Spectograms and evaluated their results

WV2VEC2 MODEL

- ✓ Load Audio Data
- ✓ Emotions Mapping

→

- ✓ Split into Train / Validation / Test
- ✓ Stratified by emotion class

→

Load Wav2Vec2 Processor (facebook/wav2vec2-base)

↓

- ✓ Load Wav2Vec2ForSequenceClassification

  → Set num classes (labels) & maps

←

- ✓ Freeze Feature Extractor Layers
- ✓ Move model to GPU
- ✓ Tokenize & Process Audio Samples using Wav2Vec2 Processor

←

TrainingArguments (batch size=4, learning rate=2e-5, epochs=8, eval strategy, Logging: Every 10 steps for progress tracking)

↓

Defind some compute metrics → Accuracy, F1, Precision, Recall (macro avg), Confusion matrix, per-class performance comparison

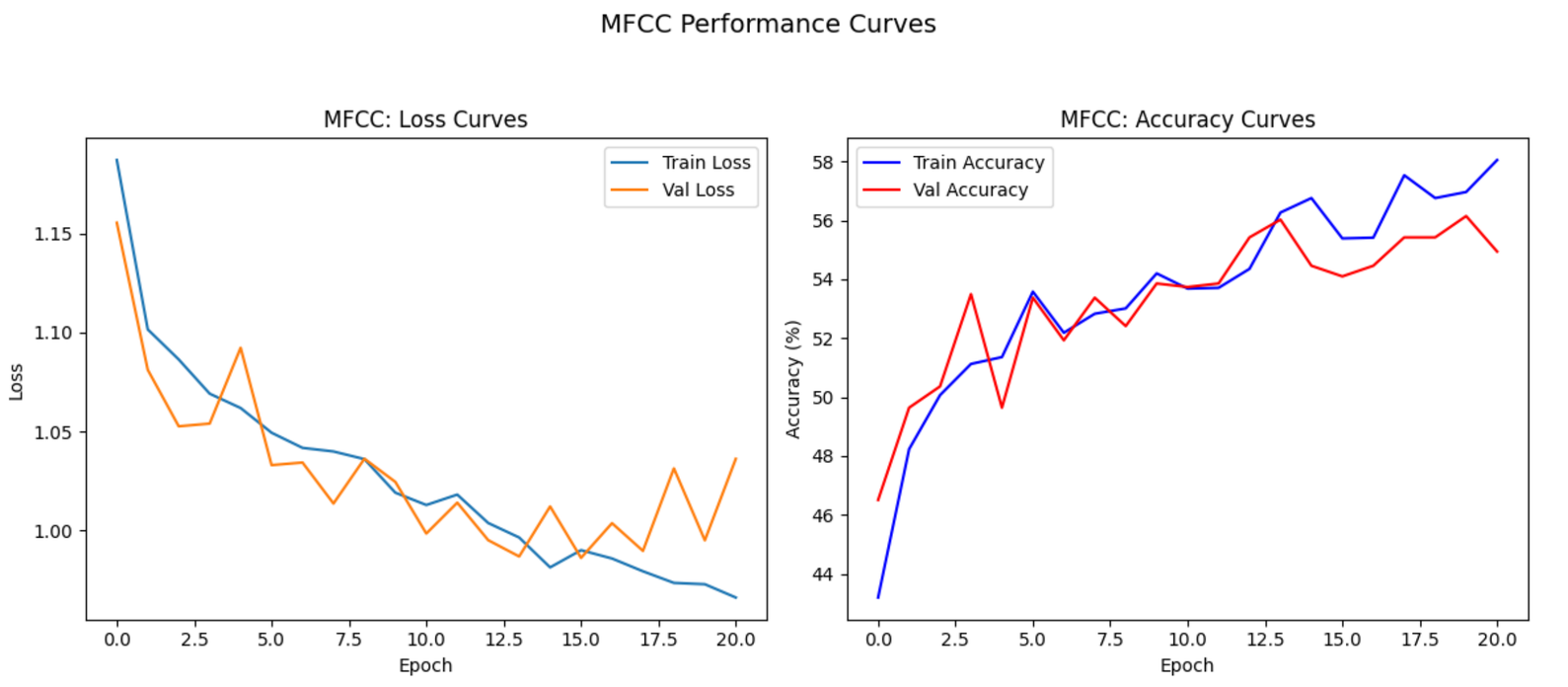
→

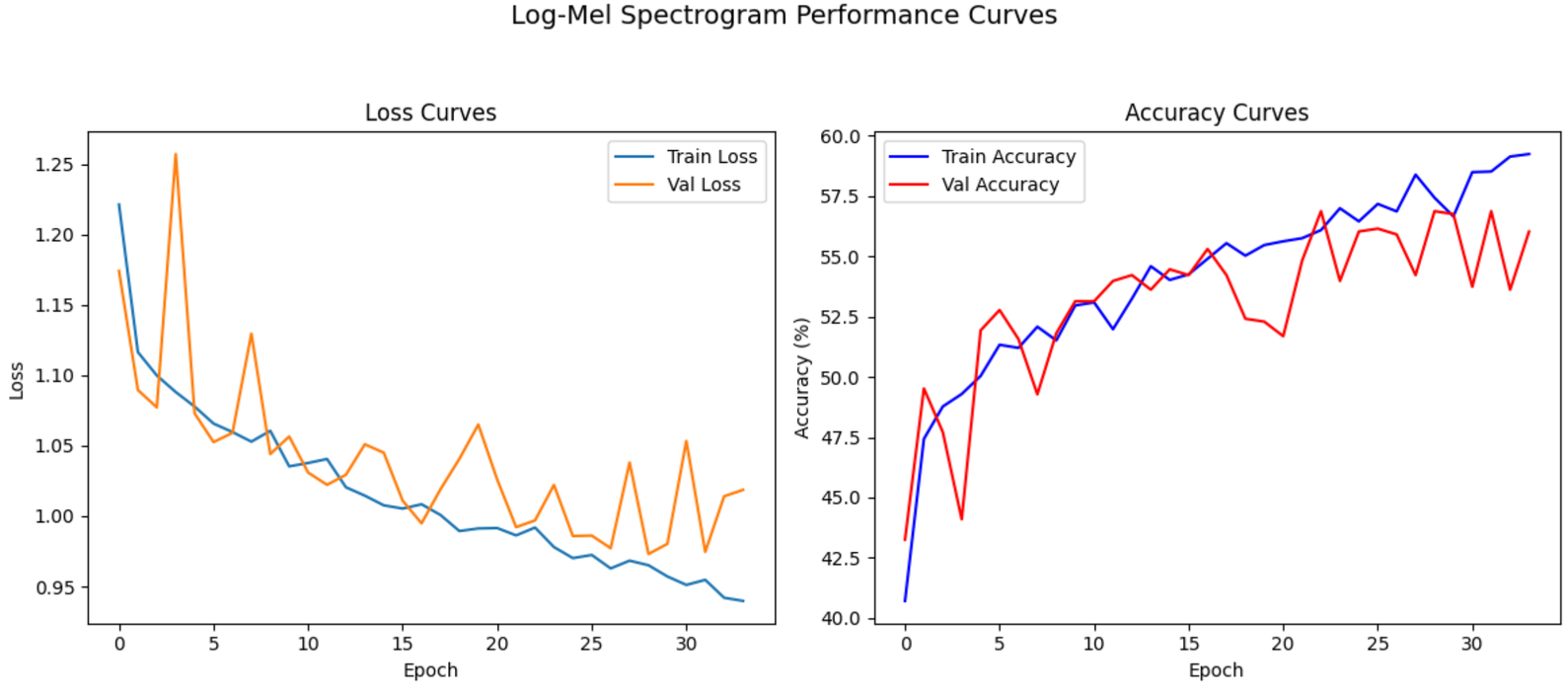
- ✓ Trainer → model + args + datasets + metrics
- ✓ Train and Validate Model

RESULT COMPARISON OF AUDIO MODALITY

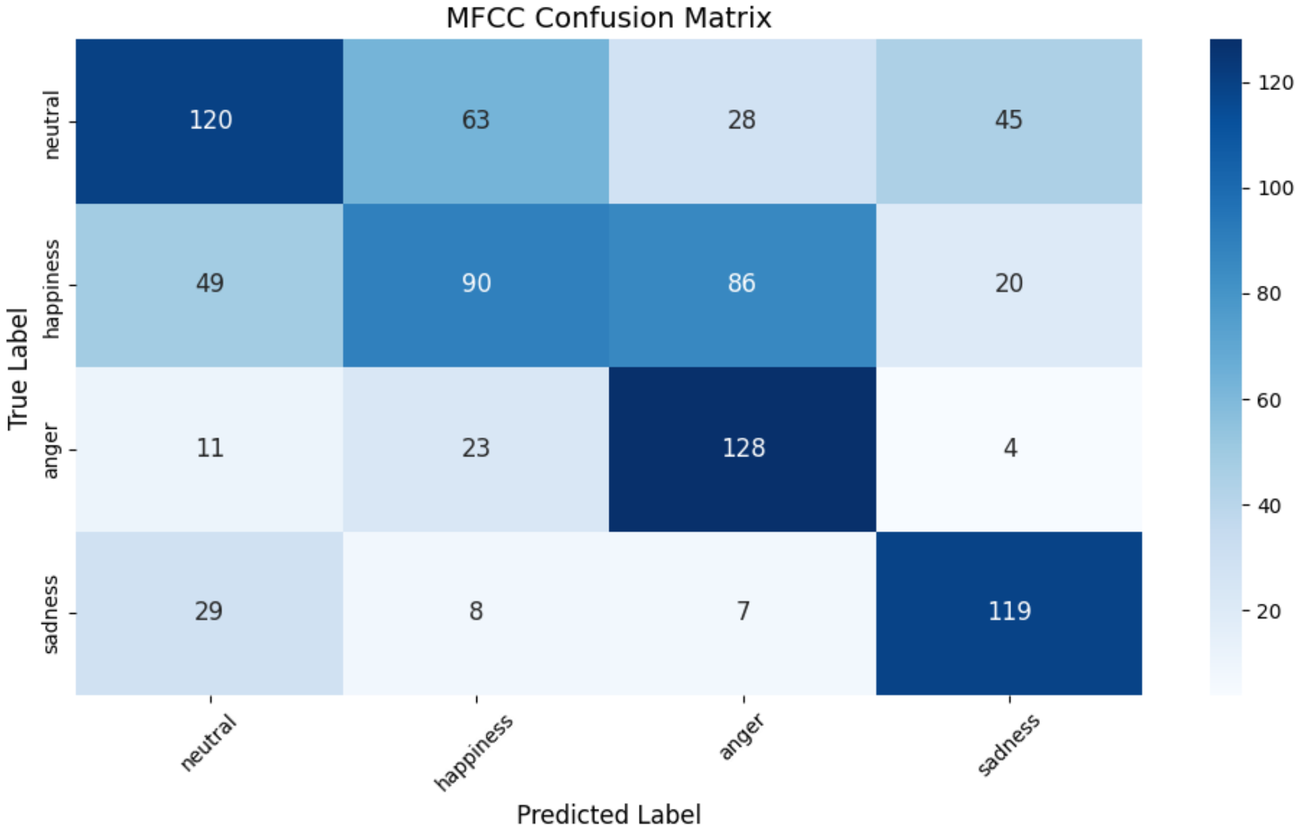
CNN

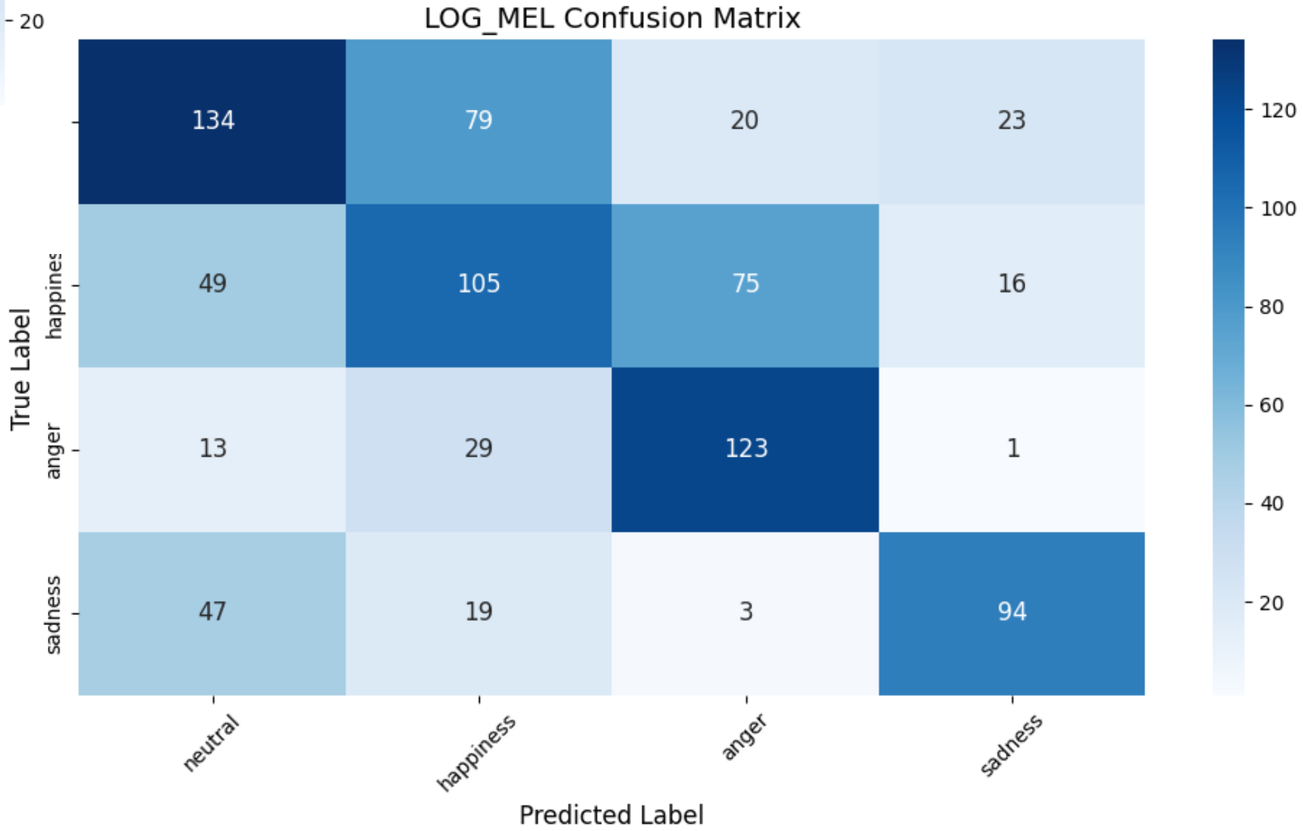
MFCC outperforms Log-Mel Spectrograms, showing higher and more stable accuracy with faster convergence and lower validation loss fluctuations.





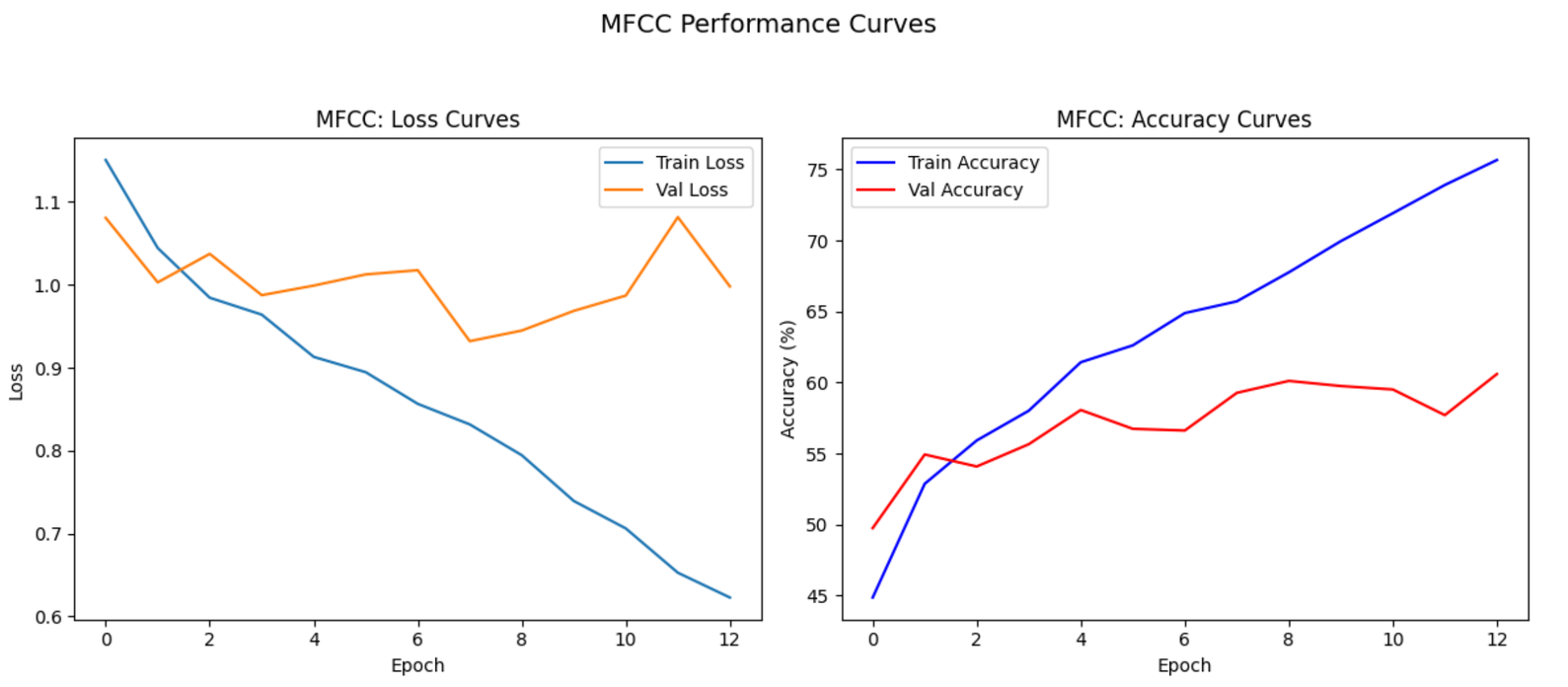
CNN

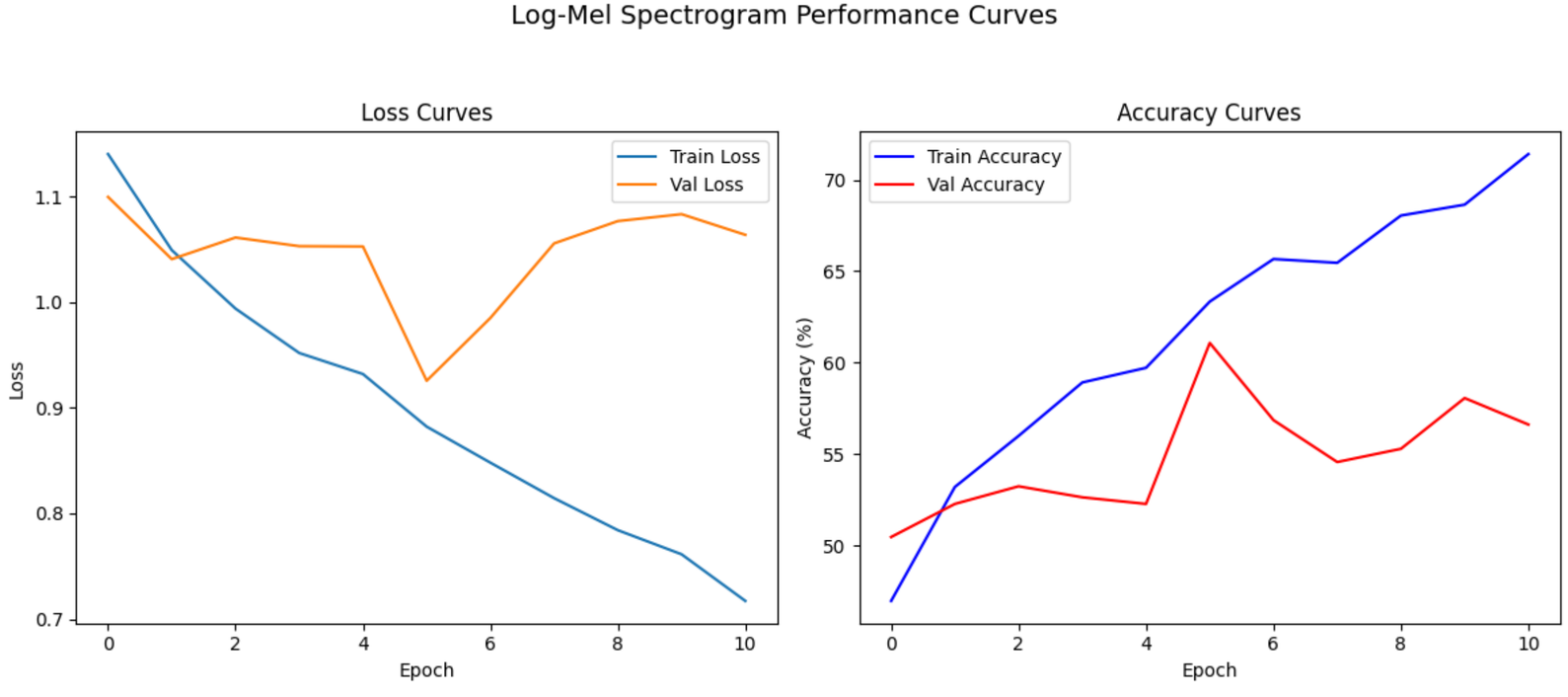

MFCC Confusion Matrix
True Label
Predicted Label
neutral
happiness
anger
sadness
120
63
28
45
49
90
86
20
11
23
128
4
29
8
7
119
120
100
80
60
40
20


LOG_MEL Confusion Matrix
True Label
Predicted Label
neutral
happiness
anger
sadness
134
79
20
23
49
105
75
16
13
29
123
1
47
19
3
94
120
100
80
60
40
20

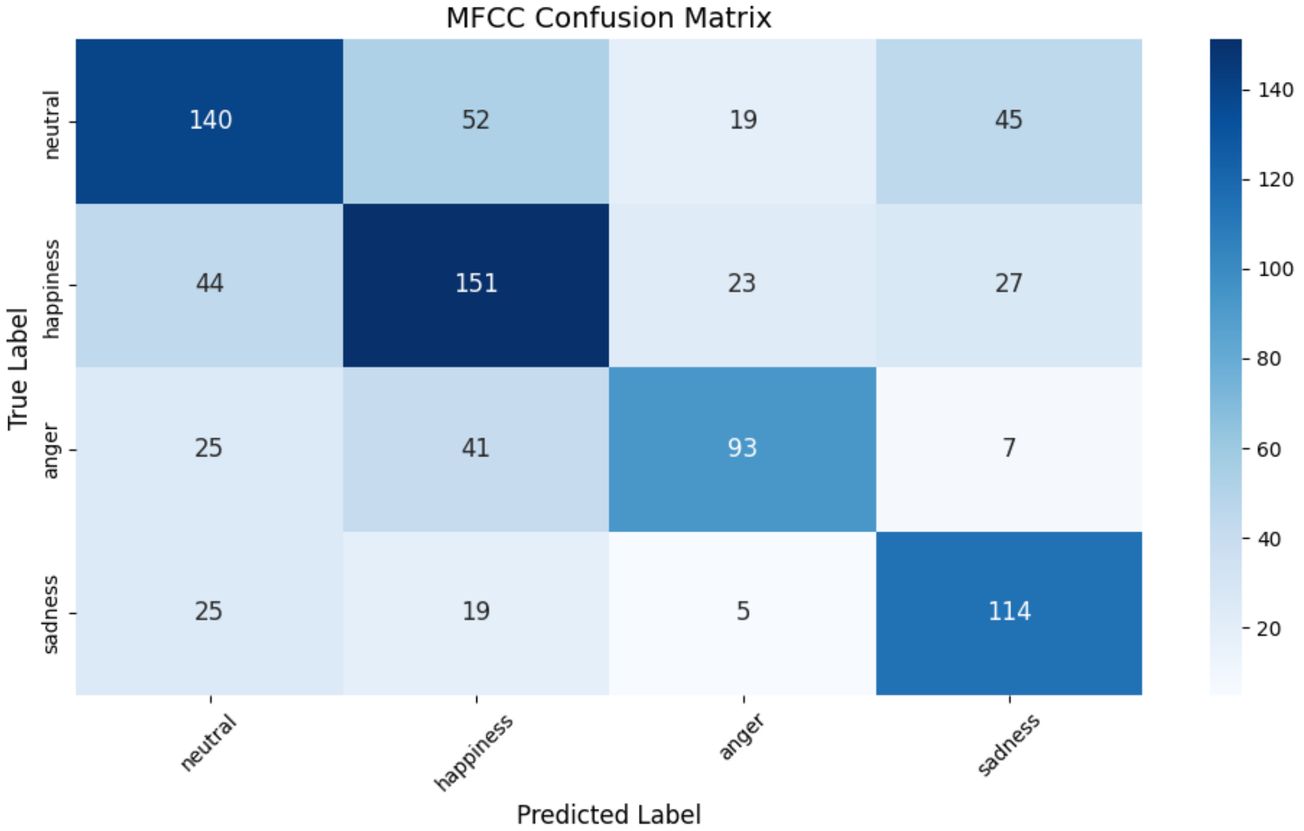
LSTM

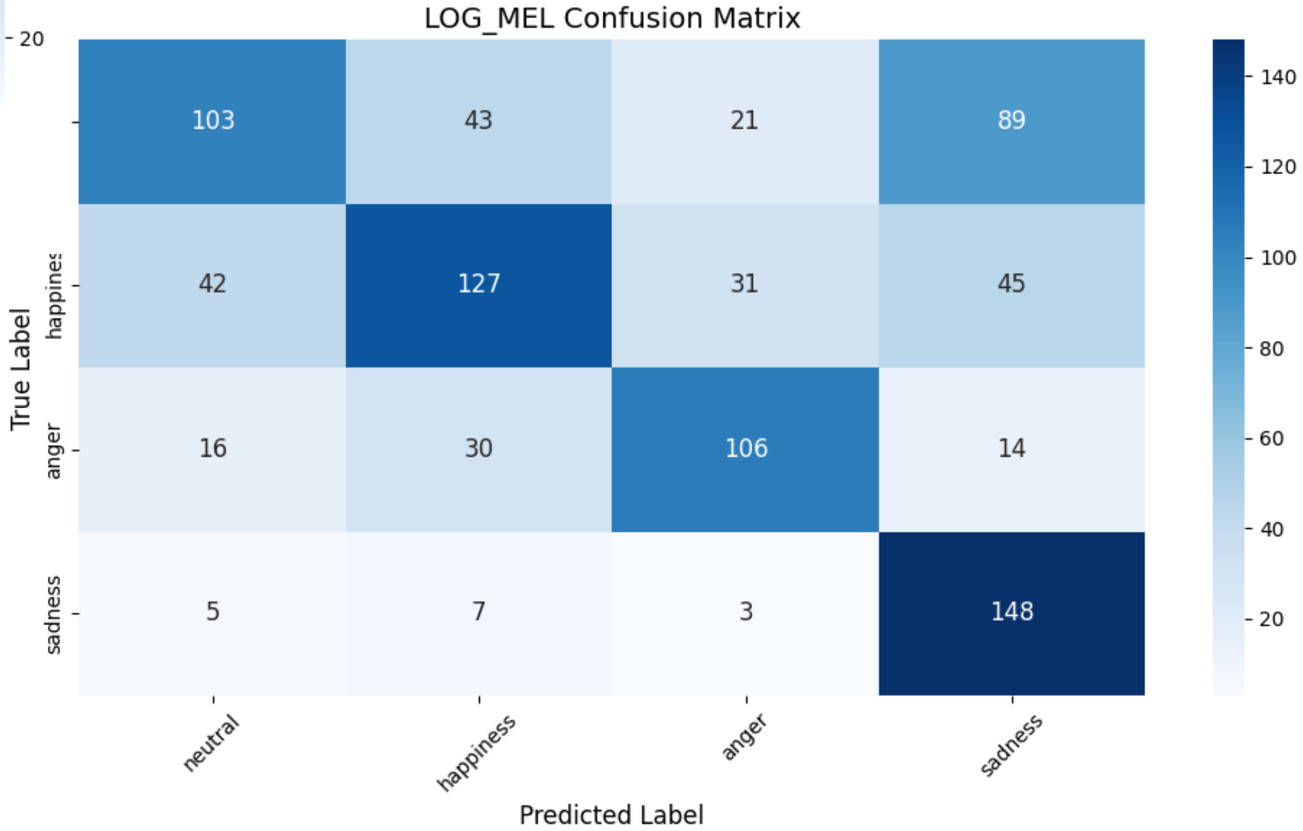
MFCCs outperform Log-Mel Spectrograms in Long Short-Term Memory (LSTM) networks, achieving higher training accuracy, even though the loss curves are moving apart from each other.





LSTM

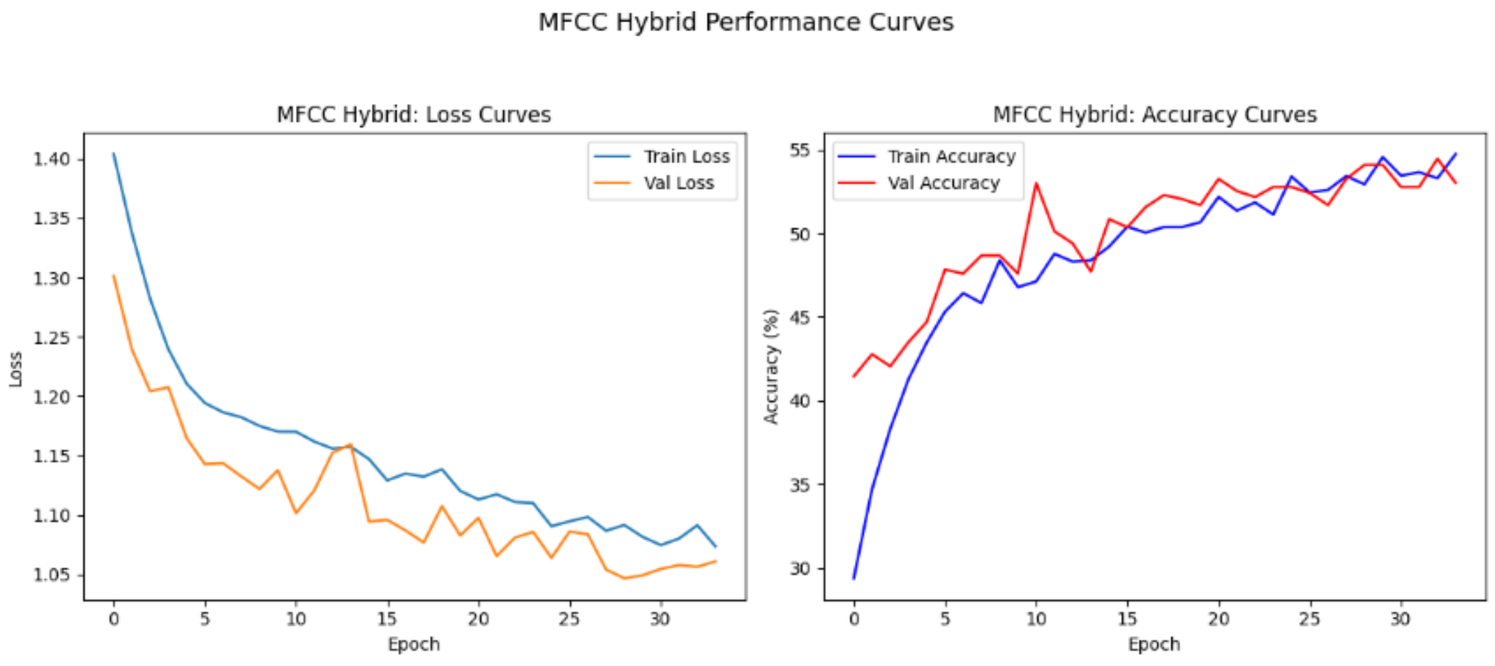


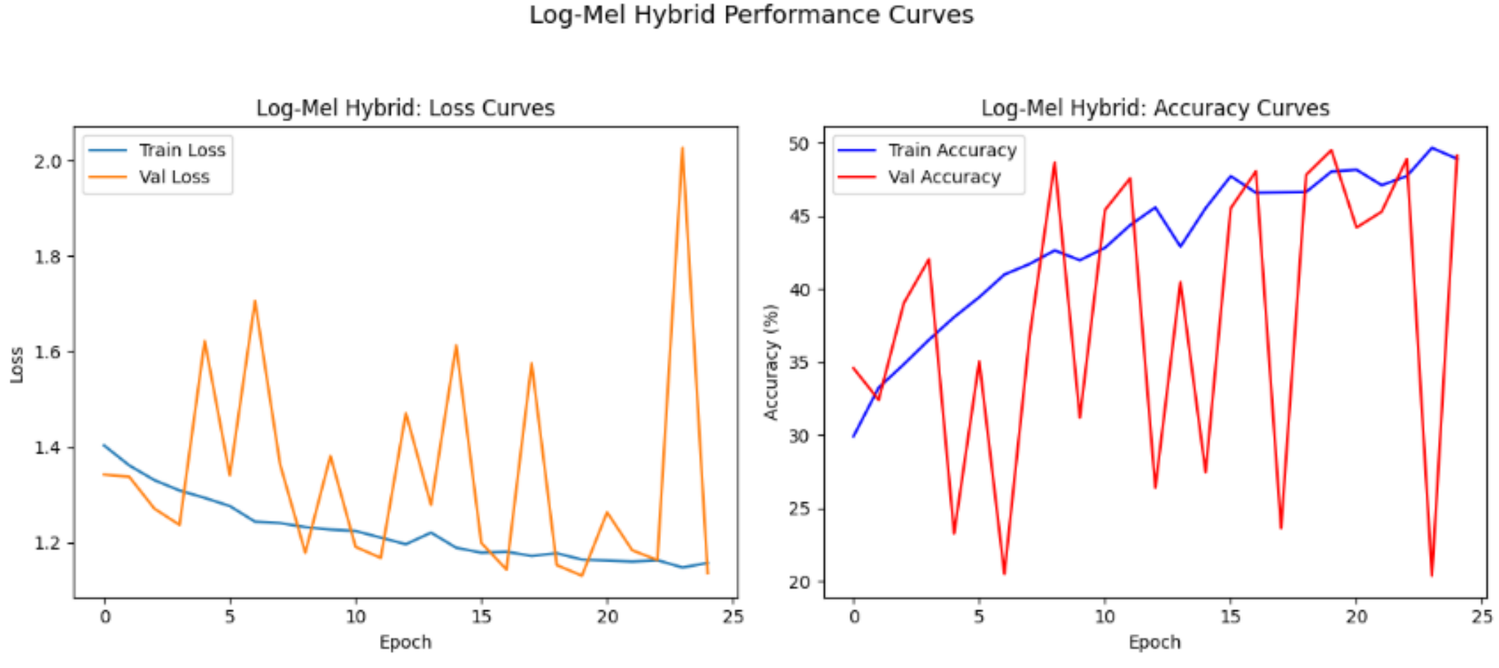


LSTM model has performed well in comparison to CNN, proving that incorporating temporal dependencies through LSTM enhances the model's ability to correctly classify audio sequences, particularly in capturing subtle patterns across time that CNNs may overlook.

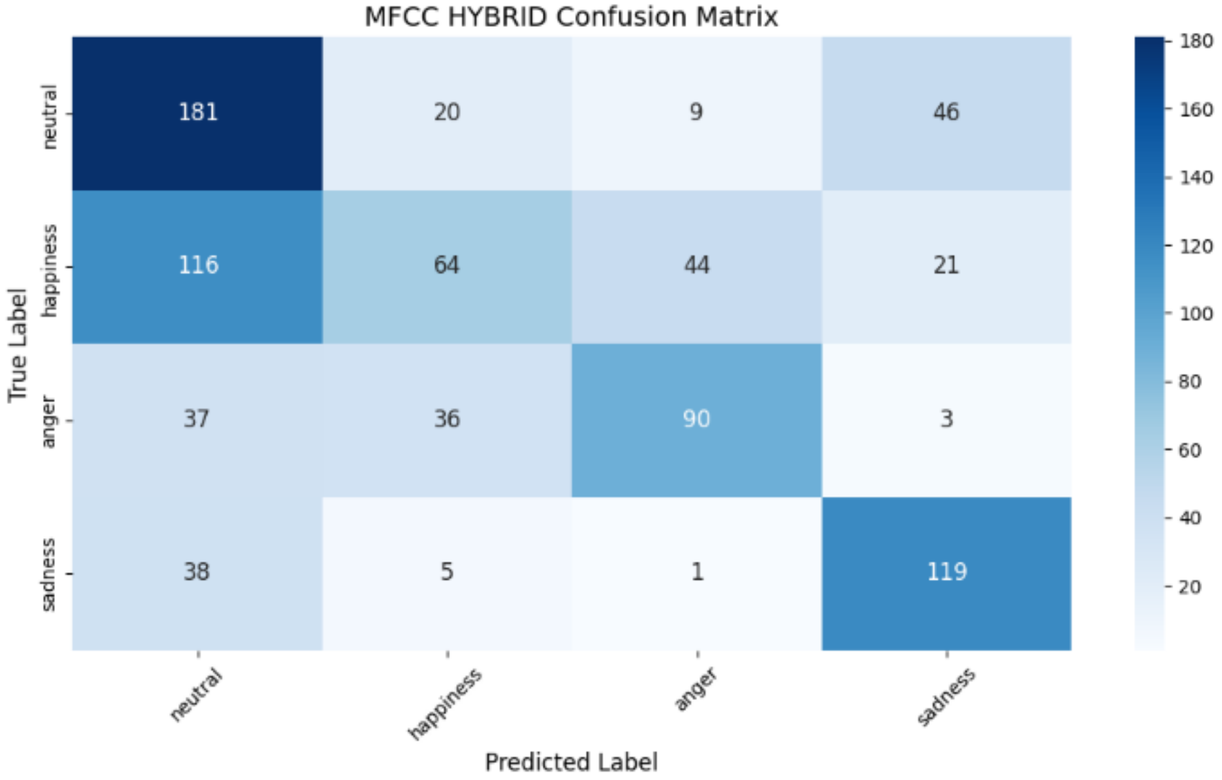
CNN-LSTM

The Log-Mel Hybrid model with Long Short-Term Memory (LSTM) shows higher peak accuracy but suffers from high validation instability, while the MFCC Hybrid model achieves more stable and consistent performance across epochs. outperforming Log-mel like previous models.





CNN-LSTM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160
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80
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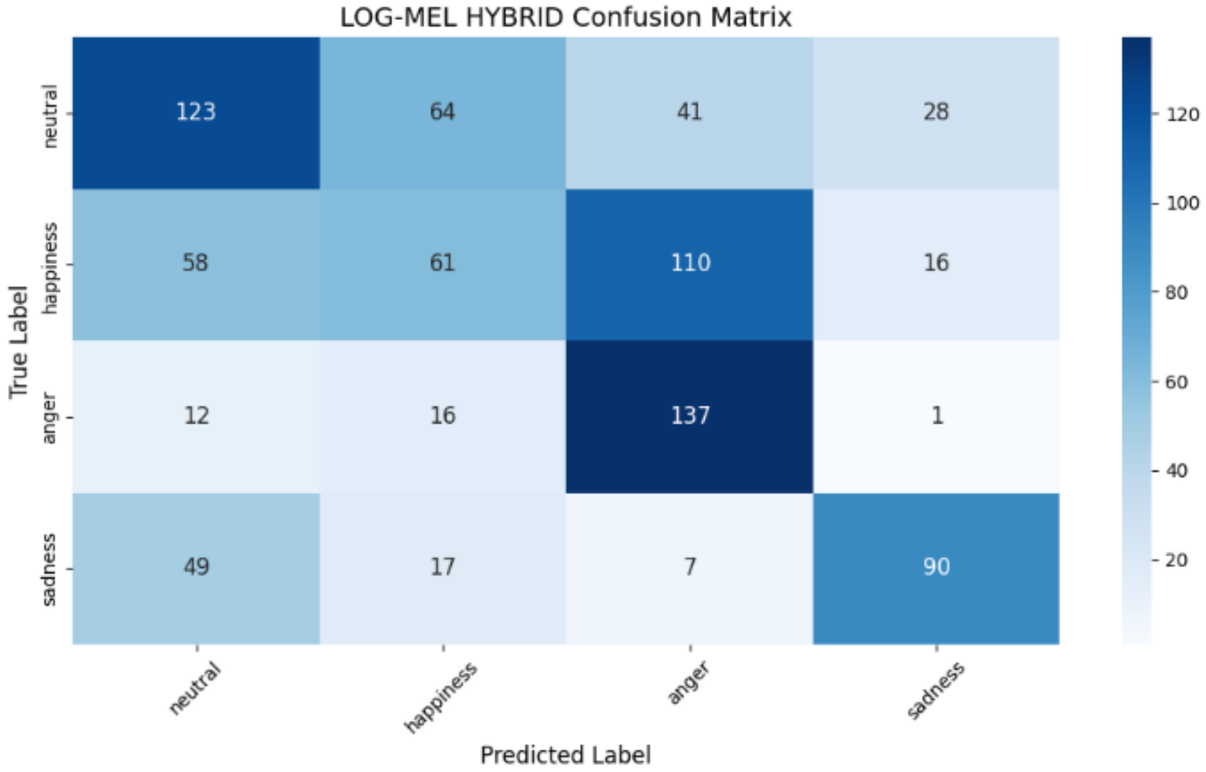
LOG-MEL HYBRID Confusion Matrix
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sadness
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61
110
16
12
16
137
1
49
17
7
90
120
100
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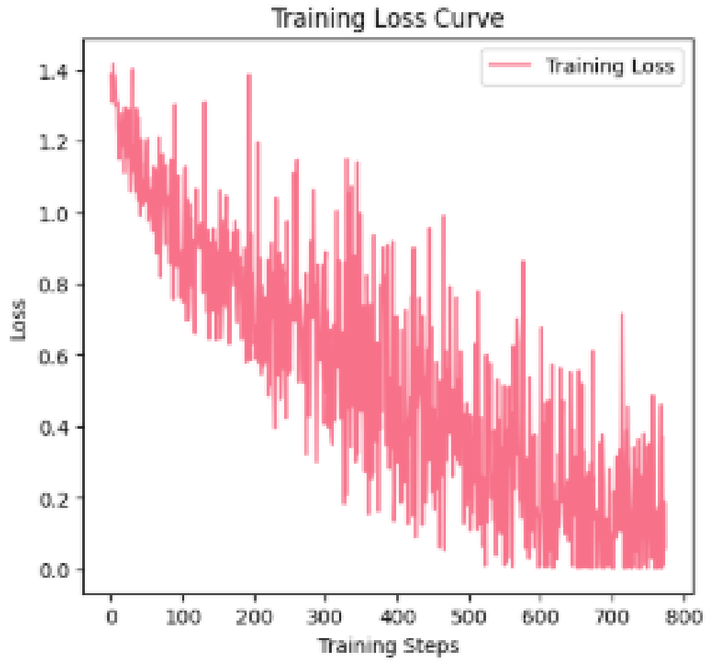
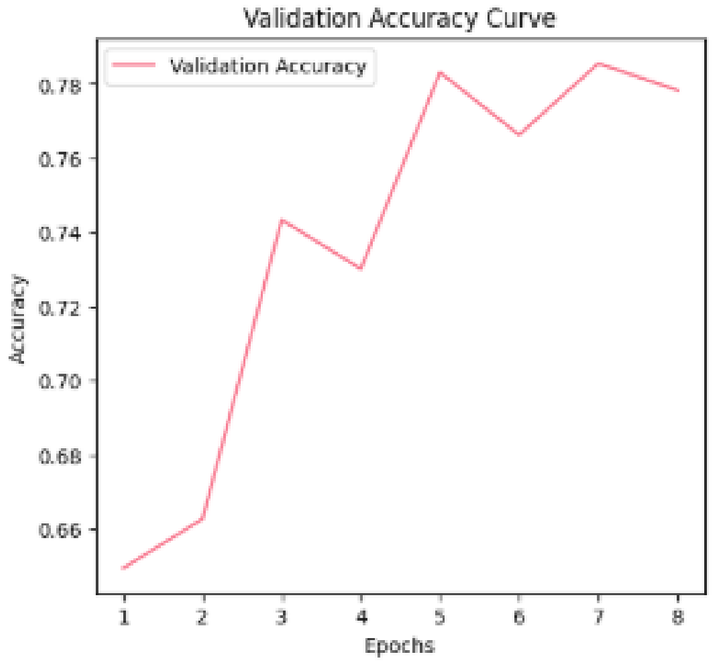
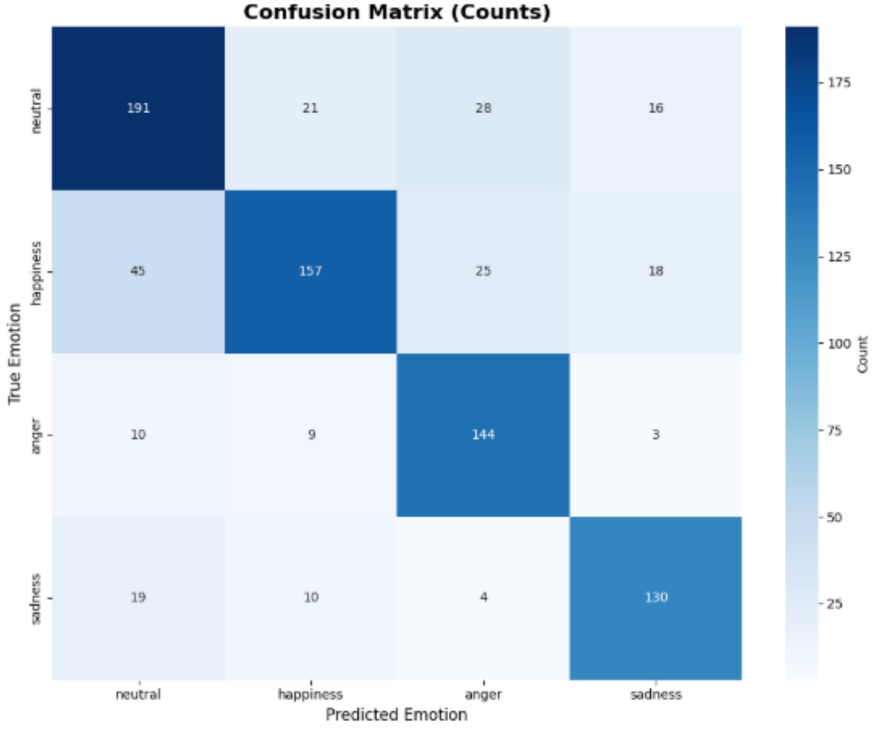
40
20

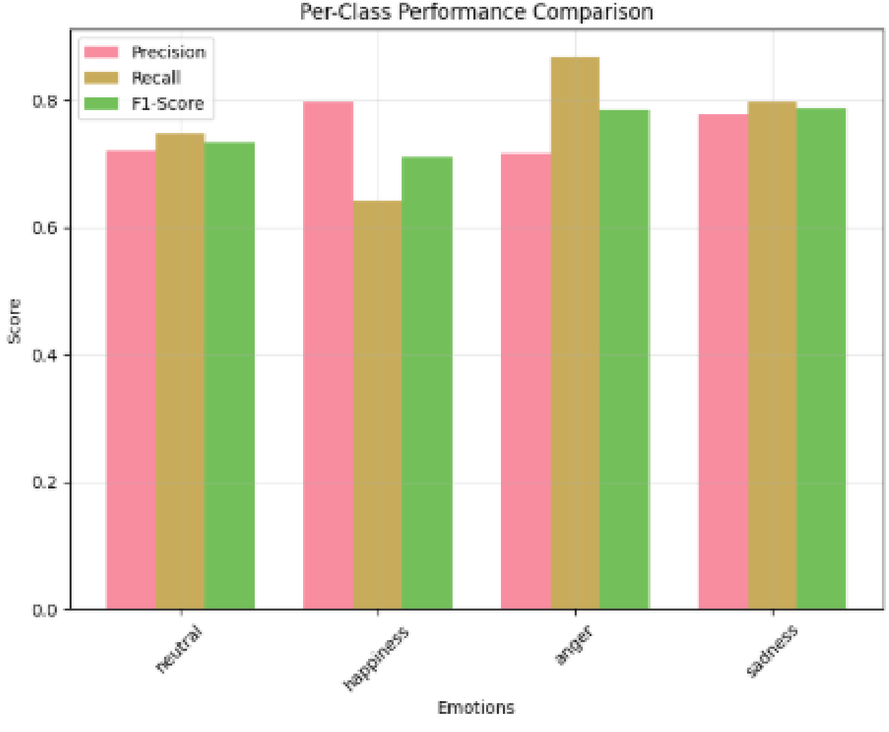
Wav2Vec2

The Wav2Vec2 model demonstrates effective learning with a consistently decreasing training loss and steadily improving validation accuracy, reaching approximately **78%**, indicating strong performance on the emotion classification task.









OVERALL AUDIO RESULT COMPARISON

Model	MFCC			Log-Mel Spectrogram		
	Accuracy	UAR	F1 score	Accuracy	UAR	F1 score
CNN	55.06%	58.43%	53.96%	54.94%	56.74%	54.70%
LSTM	60.00%	60.75%	59.91%	58.31%	57.27%	61.68%
CNN-LSTM	54.70%	56.01%	52.88%	49.52%	52.67%	47.92%

TEXT MODALITY

Classify emotions from conversational text from IEMOCAP dataset.

Audio

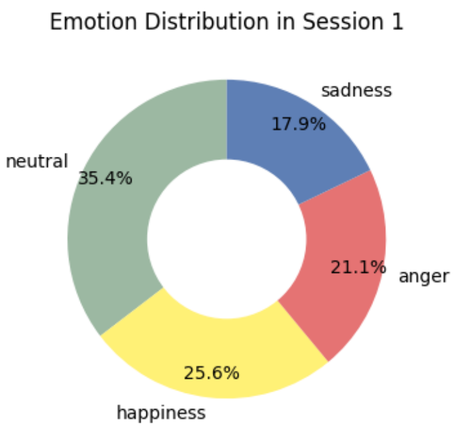
“I’m fine”

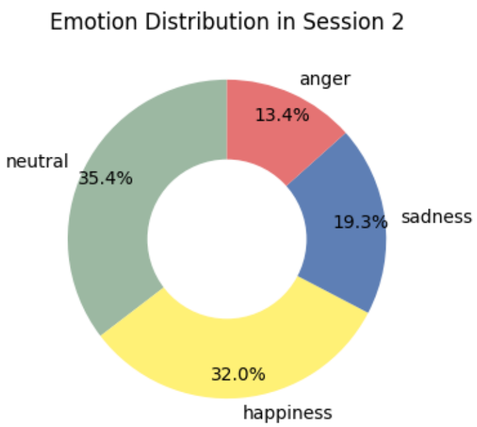
Text

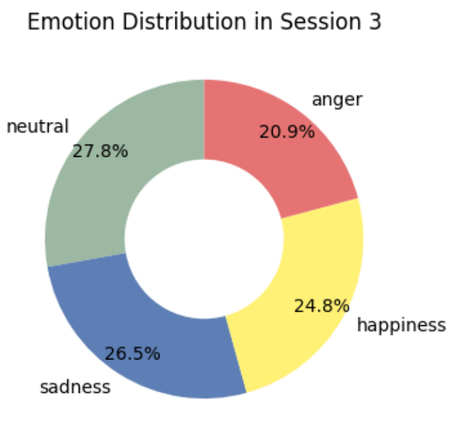
I’m fine”

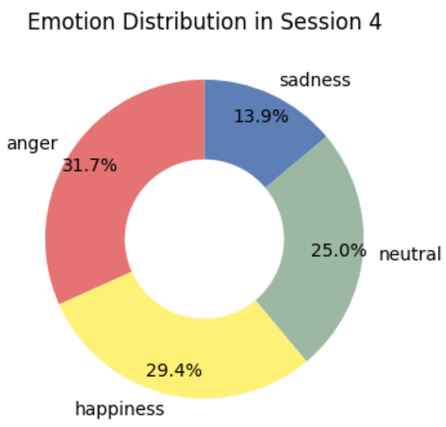
DATASET OVERVIEW

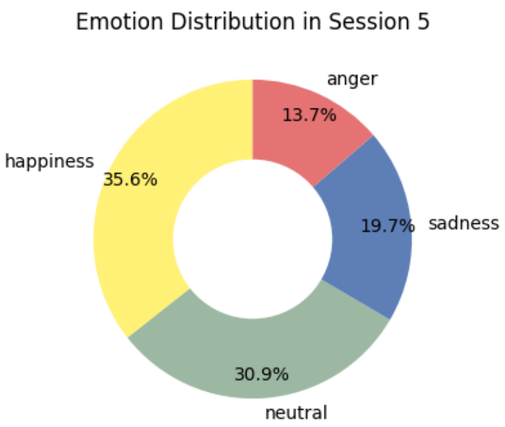
Session-based distribution



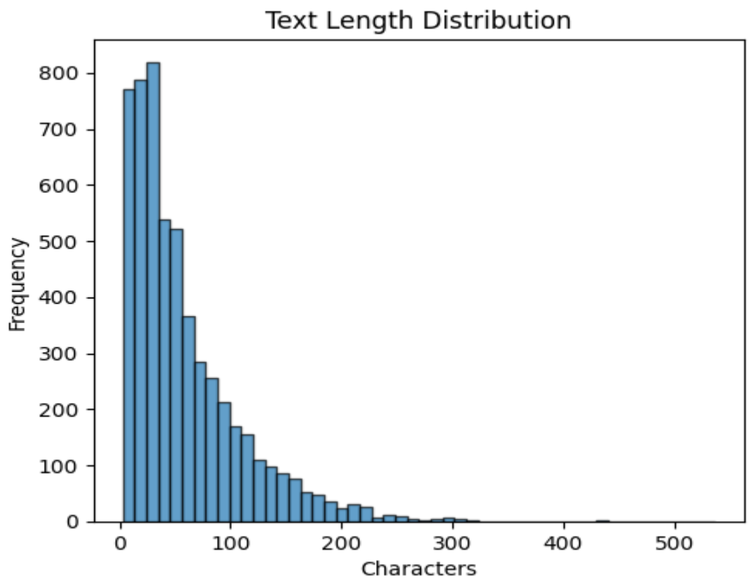


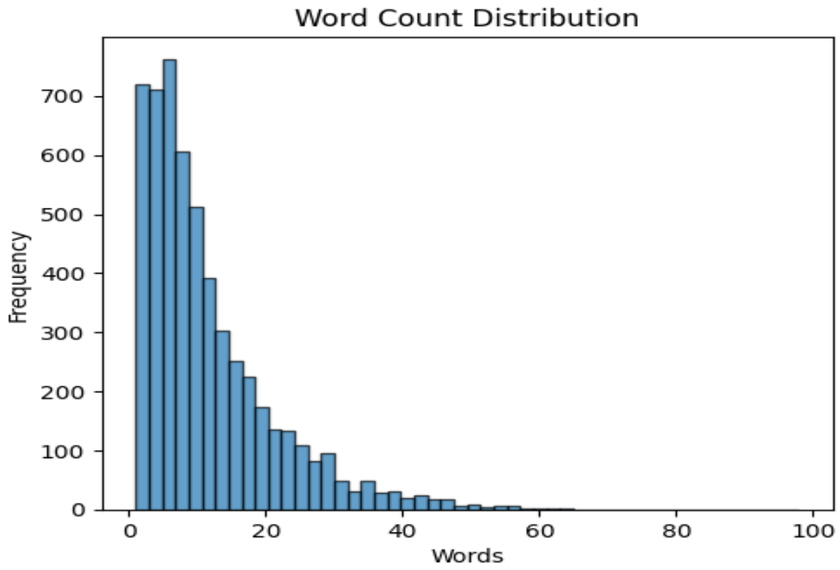


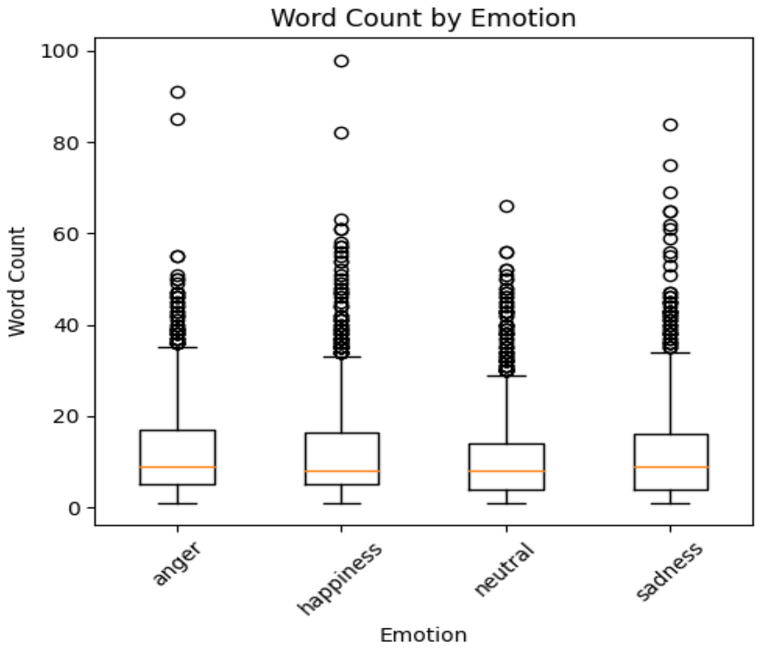




DATASET OVERVIEW







DATA PREPROCESSING

Label mapping

Merge excited with happiness, filter emotion
Target 4 Emotion:
Anger, Happiness, Sadness, Neutral

Label Encoding

- Convert string labels (“anger”, “happiness”, “sadness”, “neutral”) into numeric IDs (0, 1, 2, 3).
- Create store mapping two dictionary for model compatibility.

Id2label:(model output -> `label name`)
label2id;(`labelname` ->numerical ID)

Minimal Cleaning for BERT

- Remove dataset-specific Tags (like [LAUGH], [COUGH],[SIGH]
- Normalize Spacing & punctuation
- (hello!!! -> hello!)
- Unlike heavy preprocessing (like stemming/stopword removal), minimal cleaning was used to keep semantic richness intact.

Tokenization

- Using BERT tokenizer
- Converts text into tokens (subword units) that BERT understands.
- MAX_LENGTH=128 , chosen based on EDA -> covers 99% of utterances without wasting space on padding
- Ensure consistent input length

FEATURE EXTRACTION

BERT Embeddings
from the pre-trained model BERT-BASE-UNCASED

BERT converts each utterance into a dense vector representation that captures:

- **Meaning** (semantic content of words)
- **Syntax** (word order, grammar)
- **Emotion-related cues** (ex, emphasis, polarity of words).

MODEL DESIGN

Text → Tokenization (max_len=128) → BERT Encoder → [CLS] embedding (768-dim) → Dense Layer (768→4) → Softmax → Emotion Class

Base Architecture:

- **Foundation model:** BERT-base-uncased (Bidirectional Encoder Representations from Transformers)
- **Pre-trained Parameters**: 110 million parameters
- **Architecture**: 12 transformer encoder layers, 768 hidden dimensions, 12 attention heads.

Task-Specific Adaptations:

- **Classification Head:** Linear layer mapping BERT's pooled output ([CLS] token) to 4 emotion classes.
- **Output Layer:** 4-dimensional softmax layer for multi-class emotion classification (anger, happiness, neutral, sadness).
- **Label Mapping:** Custom id2label and label2id dictionaries for emotion categories.

TRAINING STRATEGY

Cross-Validation Approach

- **Method**: Leave-one-session-out cross-validation
- **Validation Split**: training data reserved for validation using stratified sampling to maintain class distribution

Class Imbalance Handling

- **Dynamic Class Weighting**
- **Weighted Loss Function**
- **Stratified Sampling**

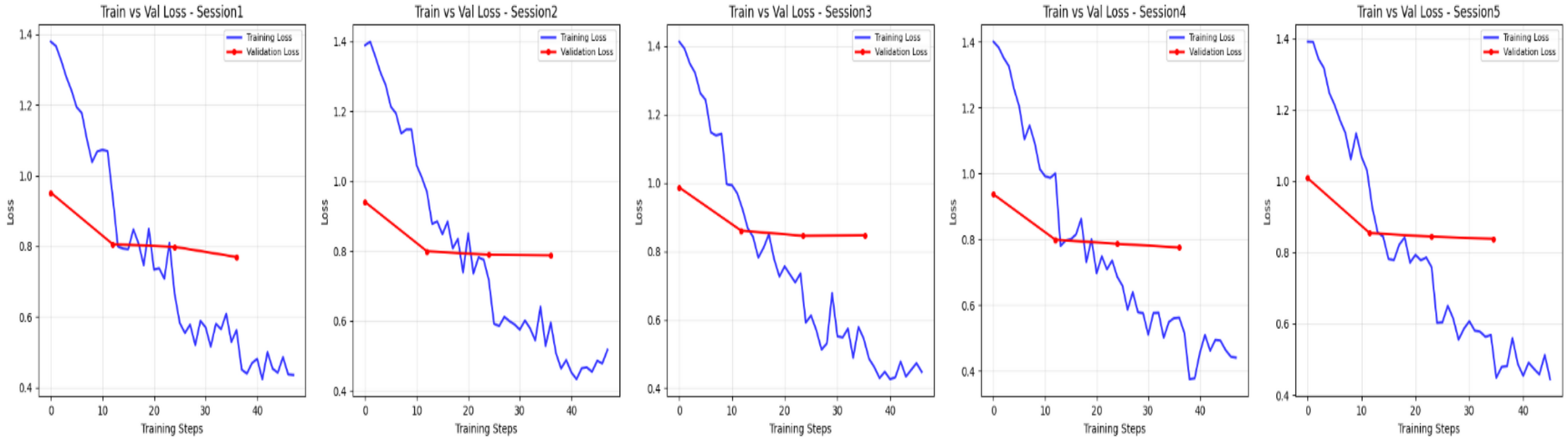
Model Training Configuration

- **Epochs**: 4 epochs
- **Early Stopping**: Patience of 3
- **Metric Selection**: Based on validation macro-F1 score
- **Batch Size**: 16 samples
- **Learning Rate**: 2e-5 with AdamW optimizer
- **Weight Decay**: 0.01
- **Reproducibility**: Fixed SEED=42

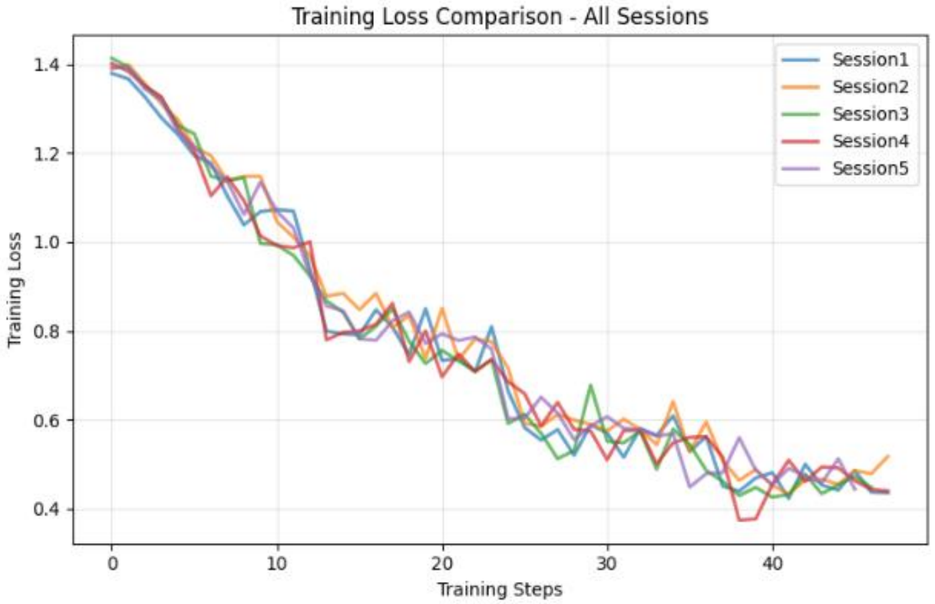
Ensures speaker-independent evaluation

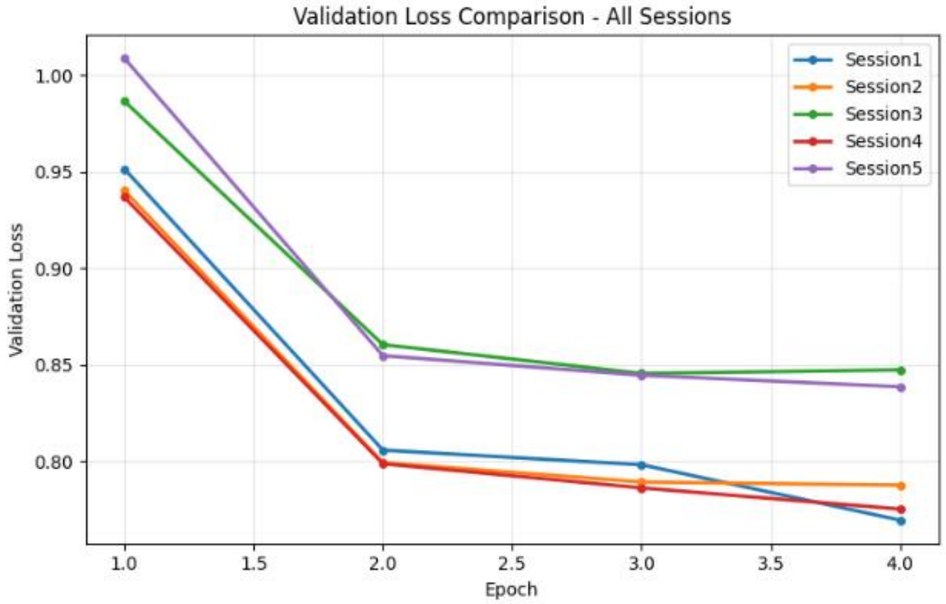
RESULT

Training vs Validation Loss (session-wise)

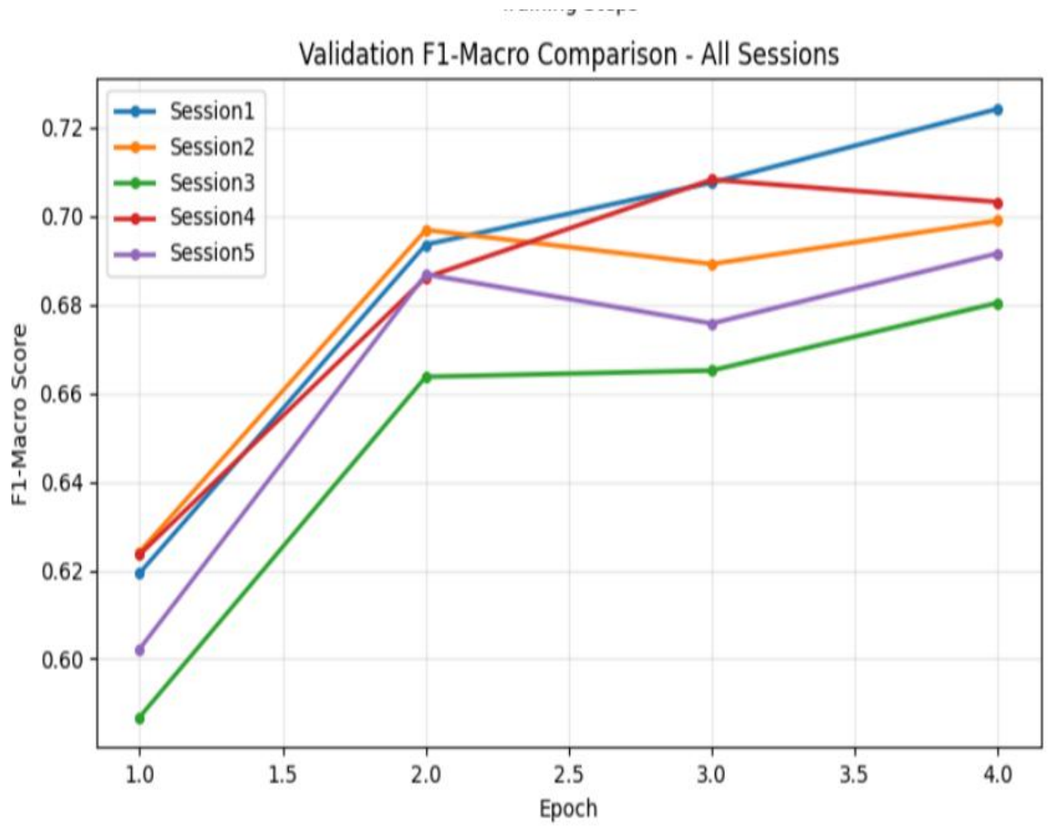


Training Loss & Validation Loss

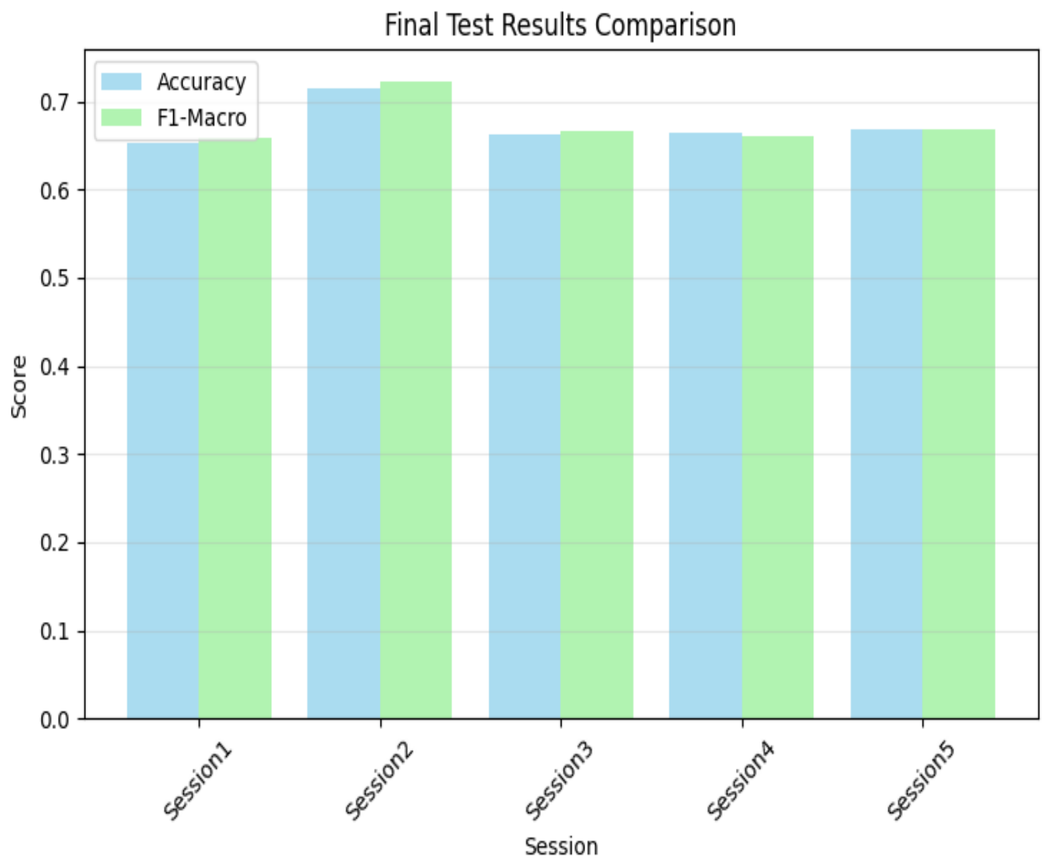




F1-Macro (All sessions)

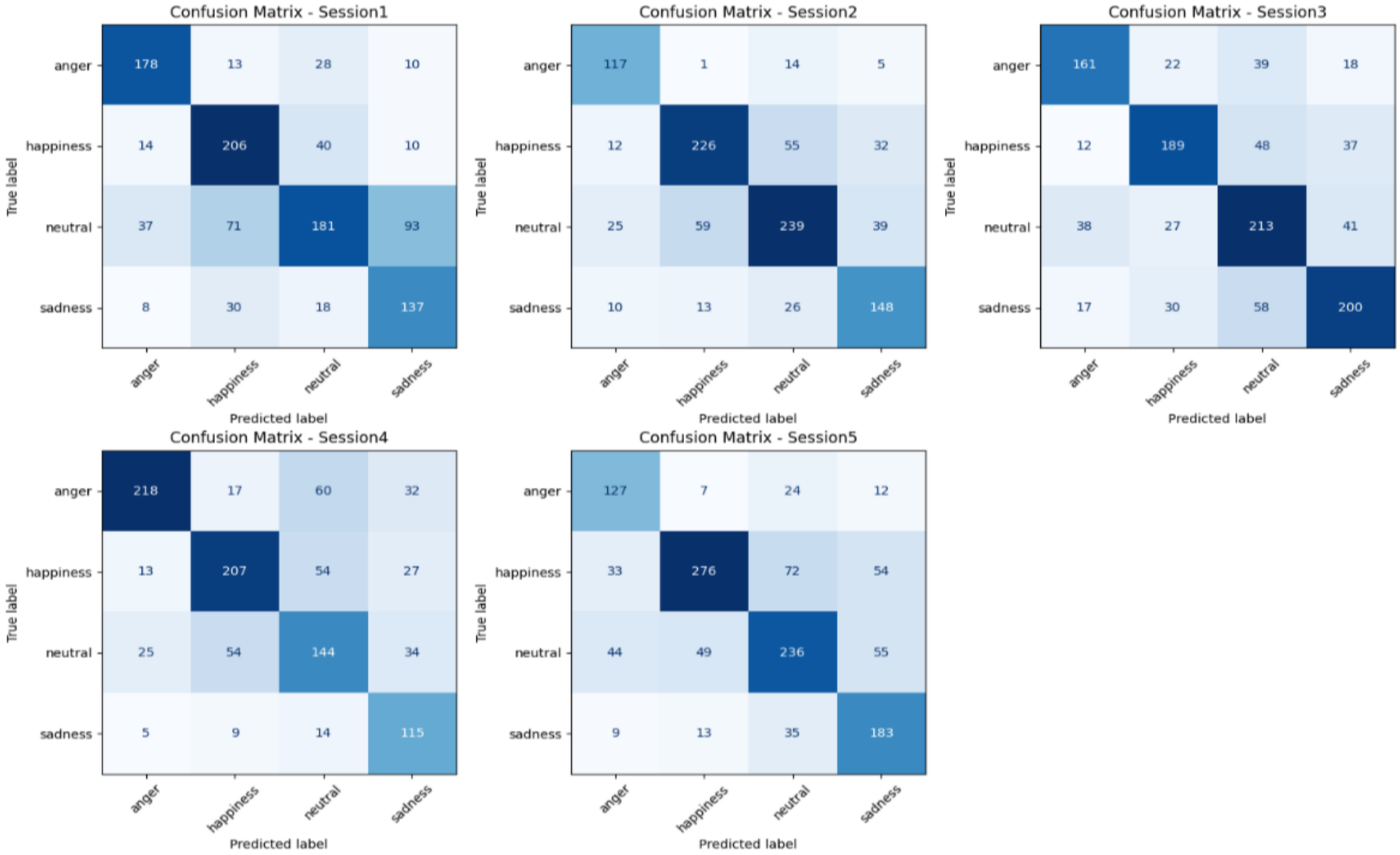


Final result comparison



Confusion Matrix of all sessions

```
Plotting confusion matrices for all sessions...
```



Overall Session-wise performance

Session	Accuracy	F1-Score	Performance
Session 1	65.4%	66.0%	Challenging (speaker-specific patterns)
Session 2	71.5%	72.3%	**Best**
Session 3	66.8%	67.2%	Moderate
Session 4	69.1%	70.1%	Good
Session 5	66.3%	67.8%	Moderate

67%
OVERALL ACCURACY

68%
Macro F1-Score

±0.03
Standard deviation

5/5
successful fold

Overall Classification report all sessions combined

Emotion	Precision	Recall	F1-score	Support
Anger	0.72	0.73	0.72	1103
Happiness	0.73	0.68	0.70	1617
Neutral	0.63	0.59	0.61	1704
Sadness	0.61	0.72	0.66	1078

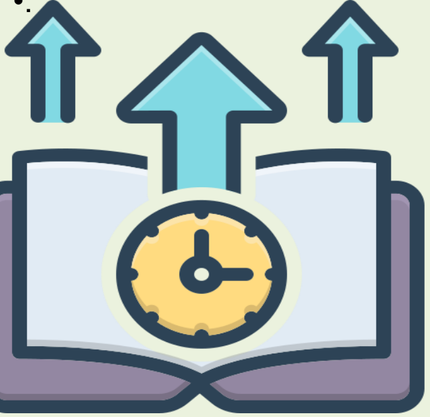
KEY INSIGHT

- Consistent performance across all sessions (66-72% range) demonstrates robust generalization to unseen speakers.
- Demonstrates BERT's effectiveness for conversational emotion classification and text modality's strengths & limitations.
- Emotion recognition from text alone is challenging because emotional cues are often subtle, context-dependent, and can be ambiguous without vocal or visual information.

→ **Solid foundation for multimodal emotion recognition**

→ **Provides baseline for audio-text fusion approaches**

→ **Highlights the importance of multimodal fusion**



IN THE NEXT SESSION

- **Multimodal Fusion**: Combine audio (MFCC/spectrogram) and text (BERT embeddings).
- **Fusion Strategies**: Evaluate early, late, and hybrid (attention-based) fusion.
- **Real-Time Deployment**: Build a user-friendly web app for speech/text input with instant emotion prediction.
- **Comprehensive Evaluation**: Compare with state-of-the-art multimodal baselines.
- **Ablation Studies**: Identify the most effective fusion approach

CONCLUSION

This research built strong unimodal baselines for SER on IEMOCAP. Wav2Vec2.0 achieved the best audio accuracy (74.9%), while BERT-based text classification averaged 67.0% across sessions. Findings show MFCC features outperform spectrograms, transfer learning is effective, and hybrid CNN-LSTM models underperform. While unimodal models are competitive, combining audio (how) and text (what) is essential for higher accuracy. Next, the project will focus on multimodal fusion and real-world deployment.